

Tetragonal-structural changes influenced the magnetic and ferroelectric properties of (Y, Fe)-codoped BaTiO₃ ceramics

Y.J. Choi^a, S.W. Kim^a, T.L. Phan^{a,b}, B.W. Lee^{a,*}, D.S. Yang^c

^a Department of Physics and Oxide Research Center, Hankuk University of Foreign Studies, Yongin, 17035, South Korea

^b Faculty of Engineering Physics and Nanotechnology, VNU-University of Engineering and Technology, 144 Xuan Thuy, Cau Giay, Hanoi, Viet Nam

^c Department of Science Education, Chungbuk National University, Cheongju, 360-763, South Korea

ARTICLE INFO

Keywords:

BaTiO₃-Related materials
Crystal/electronic structure
Magnetic/electrical properties

ABSTRACT

We present a detailed study on the structural characterization, electronic structure, and magnetic/electric-polarization properties of Ba_{1-x}Y_xTi_{1-x}Fe_xO₃ (BY_xTF_xO, $x = 0-0.1$) ceramic samples. These samples mainly crystallized in a tetragonal structure, and a small amount of a YFeO₃-related secondary phase constituted as $x \geq 0.08$. With increasing x , the lattice constant a increases while c decreases. Such structural changes reduce rapidly the intensity of characteristic Raman modes. Magnetization data have proved all BY_xTF_xO samples exhibiting ferromagnetism at room temperature. Both the saturation magnetization and coercivity tend to increase with increasing x . This is ascribed to an enhanced exchange interaction between Fe³⁺ ions, and the YFeO₃-phase formation. Herein, the Fe³⁺ presence has been confirmed upon analyzing X-ray absorption spectra. Though all the samples exhibit ferroelectricity, the electric-polarization slightly decreases with increasing x . This is associated with an increased symmetry due to the changes in the lattice parameters towards the cubic structure.

1. Introduction

Multiferroics have been known as multifunctional materials showing a co-existence of multiple properties, such as ferroelectricity, ferromagnetism and ferroelasticity [1,2]. This material system has been of intensive attention, because of potential applications in multifunctional electronic devices, and showing many interesting physical properties associated with the magnetoelectric coupling effect [3–5]. Due to large dielectric/piezoelectric responses and high thermoelectric performance, perovskite-type compounds with a general chemical formula of ABO₃ (where A and B are usually divalent earth-alkaline and transition-metal elements, respectively) have attracted much attention [6,7]. It can be fabricated ABO₃-based materials exhibiting simultaneously both ferroelectricity and ferromagnetism by doping magnetic elements into a ferroelectric host lattice [8,9], or combining ferroelectric and ferromagnetic materials in order to make composites [10–12].

Among ABO₃ compounds, BaTiO₃ (BTO) with a band-gap energy of ~3.33 eV [13] has been widely studied owing to its excellent ferroelectric and optical properties [8,14]. At room temperature, it is diamagnetic [15]. The doping of 3d-transition metals into the Ti site could induce magnetic ordering [8,9,16–20]. For example, researches on Mn-doped BTO conducted by Lin et al. [16] showed that the Mn

doping improved ferroelectric performance and magnetic behavior, while Staruch and Das et al. [19,20] found a coexistence of ferromagnetism and magnetoelectric coupling in Fe- and Ni-doped BTO. Meanwhile, a report on Co-doped BTO proved the decreased magnetodielectric and magnetoelectric properties when Co-doping content increased [21]. For the Ba-site doping, it has been investigated the influence of a rare-earth (4f) element on the multiferroic properties [22,23]. There are also some noticeable results reported on co-doped BTO compounds. It has been found an enhanced dielectric constant in (La, Ca)-doped BTO [22,24–26]. A temperature stability has been found in (Y, Ca, Zr)-codoped BTO [27]. Patel et al. [28] recorded remarkable improvements of the dielectric, ferroelectric, and optical properties in (Y, Zr)-codoped BTO at suitable doping contents. Recently, (Na, Fe)- and (Ce, Mn)-codoped BTO compounds have also been reported with many interesting characters [29,30].

For Fe-doped BTO materials, apart from a co-existence of ferroelectricity and ferromagnetism in the tetragonal structure [31], the Fe doping could promote the hexagonal phase of BTO and improve magnetic ordering [5,8,32]. Theoretical calculations have been suggested that super-exchange interactions between Fe³⁺ ions are associated with oxygen, and the exchange coupling between Fe³⁺ and Fe⁴⁺ ions leads to the multiferroic properties of Fe-doped BTO materials [33]. To get more

* Corresponding author.

E-mail address: bwlee@hufs.ac.kr (B.W. Lee).

<https://doi.org/10.1016/j.cap.2023.06.002>

Received 28 February 2023; Received in revised form 11 May 2023; Accepted 2 June 2023

Available online 7 June 2023

1567-1739/© 2023 Published by Elsevier B.V. on behalf of Korean Physical Society.

understanding about this material system, we have fabricated $\text{Ba}_{1-x}\text{Y}_x\text{Ti}_{1-x}\text{Fe}_x\text{O}_3$ and considered simultaneously the magnetic and ferroelectric properties at room temperature. The obtained results are explained upon the results obtained from analyzing X-ray powder diffraction (XRD) patterns, and X-ray absorption and Raman spectra.

2. Experimental details

In this work, $\text{Ba}_{1-x}\text{Y}_x\text{Ti}_{1-x}\text{Fe}_x\text{O}_3$ (hereafter, named $\text{BY}_x\text{TF}_x\text{O}$) samples with $x = 0, 0.01, 0.02, 0.04, 0.08$, and 0.1 were prepared by a conventional solid-state reaction method in air. BaCO_3 (Sigma-Aldrich, 99.5%), TiO_2 (Sigma-Aldrich, 99.8%), Y_2O_3 (Sigma-Aldrich, 99.99%), and Fe_2O_3 (Alfa-Aesar, 99.9%) were used as initial chemicals to fabricate the samples. These chemicals were scaled according to stoichiometric ratios of elements present in $\text{Ba}_{1-x}\text{Y}_x\text{Ti}_{1-x}\text{Fe}_x\text{O}_3$ compositions. They were well ground, and then pre-annealed at 800°C for 6 h. After pre-annealing, they were re-ground and pelletized with the addition of 3–5 wt% of polyvinyl alcohol and conducted a first sintering step at 1000°C for 6 h. The final sintering step was conducted at 1350°C for 6 h after the mixtures were ground and pelletized in pellets once again. Final products in pellets were measured the apparent density (d_A). Its value ranges between 3.22 and 4.21 g/cm^3 for $x = 0$ – 0.1 , as shown in Table 1. The surface morphology was checked by a scanning electron microscope (SEM, Jeol JSM-7600F). The crystal structure of the samples was checked by an X-ray diffractometer (Miniflex, Rigaku) equipped with the $\text{Cu-K}\alpha_1$ radiation source, and a micro-Raman spectrometer (XploRA PLUS, Horiba) using a laser wavelength of $\lambda_{\text{exc.}} = 532\text{ nm}$. X-ray absorption spectroscopy (XAS, Pohang Accelerator Laboratory) was used to identify the valence of Fe. The magnetic properties were investigated by a vibrating sample magnetometer (VSM 736, Lake Shore Cryotronics), where the magnetic field (H) could increase up to 10 kOe . Electric polarization (P - E) measurements were conducted on a ferroelectric tester (Precision RT66B, Radiant) using a high-voltage amplifier (609E-6, Trek).

3. Results and discussion

Firstly, we have checked the surface morphology of all fabricated $\text{BY}_x\text{TF}_x\text{O}$ ($x = 0$ – 0.1) samples. The SEM micrographs shown in Fig. 1 (a–f) indicate that the sample pellets consist of agglomerated particles, which have irregular shapes and arbitrary orientation. Their morphology remains less changed versus doping content x . The size distribution is wide, and ranges between about 0.2 and $1\text{ }\mu\text{m}$. The agglomeration of microparticles in large clusters also created a lot of micro-sized gaps. This results in low d_A values (3.22 – 4.21 g/cm^3) of the samples, as shown in Table 1. It should be noticed that microstructural features in the samples are somewhat different according to x , which could be related to the pelletizing process as using a hydraulic press, and the annealing at temperatures lower than melting points of the samples.

Together with considering the surface morphology, the crystalline structure of $\text{BY}_x\text{TF}_x\text{O}$ has been checked by an X-ray diffractometer, which swept in the angle range of $2\theta = 20$ – 70° . Fig. 2 shows XRD patterns of the samples. The patterns consist of strong and sharp peaks, indicating good crystal quality. Some twin peaks observed at $45.8, 56.9$,

and 66.6° for $x = 0$ corresponding to tetragonal BTO [34] become gradually overlapped when x increases. This demonstrates a crystal-structural change induced by (Y, Fe)-codoping. Based on the Rietveld refinement using a structural model for tetragonal BTO (space group: $P4mm$) with the initial values of the lattice constants $a = b = 3.994\text{ }\text{\AA}$ and $c = 4.038\text{ }\text{\AA}$ [35], we have found a good match between the experimental data of $\text{BY}_x\text{TF}_x\text{O}$ and the theoretical model, with good-of-fit (R_B – Bragg, and R_f – profile) factors less than 10, as shown in Table 1 and Fig. 2. Apart from the main tetragonal structure, the samples with $x \geq 0.08$ have the addition of a YFeO_3 -related secondary phase (its volume fraction is less than 5%), which corresponds to a strong XRD peak at about 33.9° .

To explain the change of twin peaks, it is necessary to consider the variation tendency of the lattice constants, a and c . The data shown in Fig. 3 exhibit that a gradually increases while c decreases when x increases. The results also indicate a gradual reduction of the c/a ratio towards the unity of a cubic structure, but no tetragonal-cubic structure transition, as shown in the inset of Fig. 3. For the tetragonal structure of BTO, it is known that a and c are the Ba–Ba and $2 \times (\text{Ti–O}_1)$ bond distances, respectively [36]. Thus, the change in the lattice constants is related to two factors: the difference in the ionic radius (r_{ion}) of dopants, and the change in electrostatic repulsion between the doped ions (Y^{3+} and Fe^{3+}) in comparison to the replaced ions (Ba^{2+} and Ti^{4+}) of the host lattice; the existence of Fe^{3+} will be presented below. Thus, the substitution of Y^{3+} ($r_{\text{ion}} = 0.9\text{ }\text{\AA}$) and Fe^{3+} ($r_{\text{ion}} = 0.645\text{ }\text{\AA}$, for high spin) for Ba^{2+} ($r_{\text{ion}} = 1.35\text{ }\text{\AA}$) and Ti^{4+} ($r_{\text{ion}} = 0.605\text{ }\text{\AA}$) [37], respectively, caused the shrinking of c and elongating of a . Having investigated (Bi, Mn)-doped BTO [38], it was also observed the same variation tendency of a and c when x increases. In crystallography, this structural modification would overlap twin XRD peaks, and influence phonon vibration modes, as shown below. Studying Sr-, Bi-, Mn-, Fe-, and Co-doped BTO materials, it was usually found a tetragonal-hexagonal transformation when doping concentration increases [8,21,39,40]. Meanwhile, it was observed the transformation from tetragonal through cubic to rhombohedral structure in (Bi, Fe)-doped BTO [41], the tetragonal-orthorhombic transition in (Gd, Fe)-codoped BTO [42], and the tetragonal space-group transition of $P4mm$ - $P4/mmm$ [38]. It is necessary to add that pure BTO can stay at different structural phases (such as cubic, tetragonal, orthorhombic, rhombohedral, and monoclinic) as changing temperature [36,43].

We have also calculated the X-ray density (d_X) using the data of the lattice constants and the expression $d_X = ZM/N_A V$, where Z ($= 3$) is the number of molecules per unit cell, and M , N_A and V are the molecular weight, Avogadro's number, and the unit-cell volume, respectively. The data shown in Table 1 indicate a gradual decrease of d_X from 5.97 g/cm^3 for $x = 0$ – 5.88 g/cm^3 for $x = 0.1$. It means that the relative-density change (d_X/d_A) according to x is about 1.42 – 1.84 , as shown in Table 1. The replacement of Y (98 g) and Fe (56 g) with the smaller total atomic mass for Ba (137 g) and Ti (48 g) with a larger total atomic mass is assigned to the reduction of d_X as observed. If combining the data of d_A and d_X , it is possible to calculate the porosity (p) as using the expression $p = (1 - d_A/d_X) \times 100\%$. As shown in Table 1, the porosity of fabricated $\text{BY}_x\text{TF}_x\text{O}$ ceramics is fairly large, ranging between 29.6 and 45.7% . This is due to a low value of d_A and a large size of gaps between microparticles (as shown in the SEM images), which are usually influenced by sample annealing conditions.

Notably, as explaining the change in lattice parameters a and c , we have mentioned Fe^{3+} ion. Its presence in $\text{BY}_x\text{TF}_x\text{O}$ could be affirmed upon analyzing X-ray absorption near edge structure (XANES) spectra [44]. Herein, we have selected two typical samples with $x = 0.04$ and 0.1 (where Fe contents are large enough to record good absorption signals), and LaFeO_3 was used as reference. Their XAS spectra in the energy range $E = 7110$ – 7200 eV are shown in Fig. 4(a). In the XANES region of 7120 – 7130 eV , there is a rapid signal-intensity change, which is assigned to the K absorption edge of Fe. Performing the derivative spectra, as shown in Fig. 4(b), one can see the first peak at $\sim 7125\text{ eV}$ of

Table 1

Experimental values obtained as analyzing the physical properties of typical $\text{BY}_x\text{TF}_x\text{O}$ ceramics.

x	d_A (g/cm ³)	d_X (g/cm ³)	d_X/d_A	R_B	R_f	P (%)	P_{max} ($\mu\text{C}/\text{cm}^2$)	E_c (kV/cm)
0	4.21	5.97	1.42	8.32	5.70	29.6	0.051	16.03
0.02	3.35	5.95	1.66	7.62	5.26	43.7	0.049	15.06
0.04	3.22	5.92	1.77	7.71	5.49	45.7	0.049	13.16
0.08	3.28	5.88	1.84	5.92	4.19	44.2	0.048	13.22
0.1	3.38	5.87	1.79	5.57	4.25	42.5	0.045	15.88

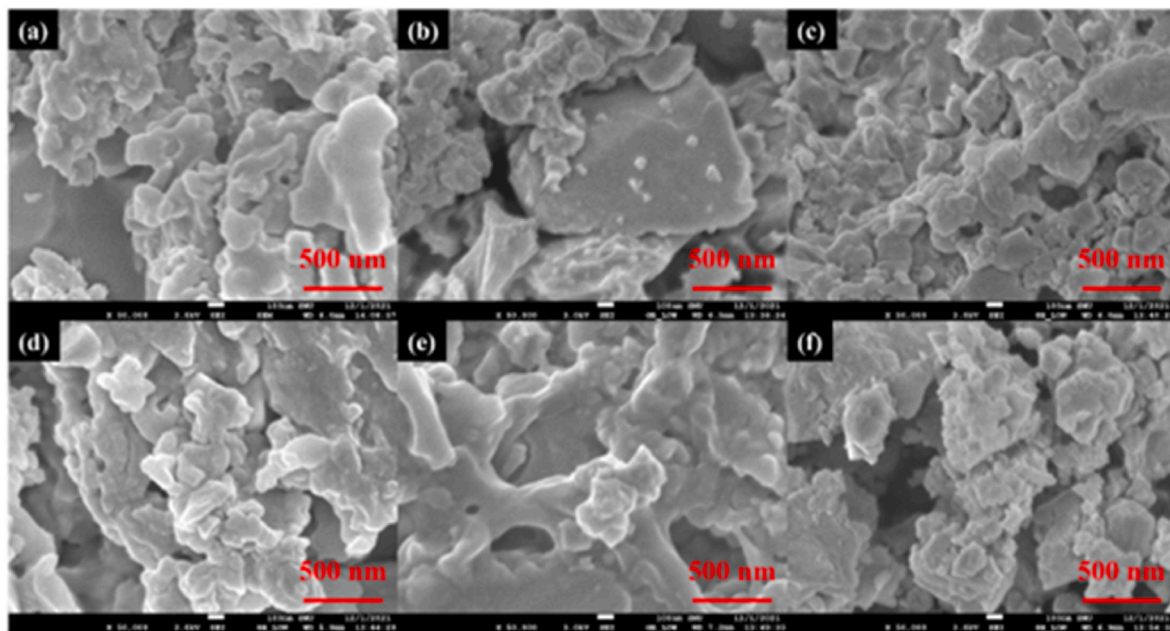


Fig. 1. SEM images showing surface morphologies of all as-prepared $\text{BY}_x\text{TF}_x\text{O}$ pellets: (a) $x = 0$, (b) $x = 0.01$, (c) $x = 0.02$, (d) $x = 0.04$, (e) $x = 0.08$, and (f) $x = 0.1$, which were recorded under the same magnification, corresponding a bar scale of 500 nm.

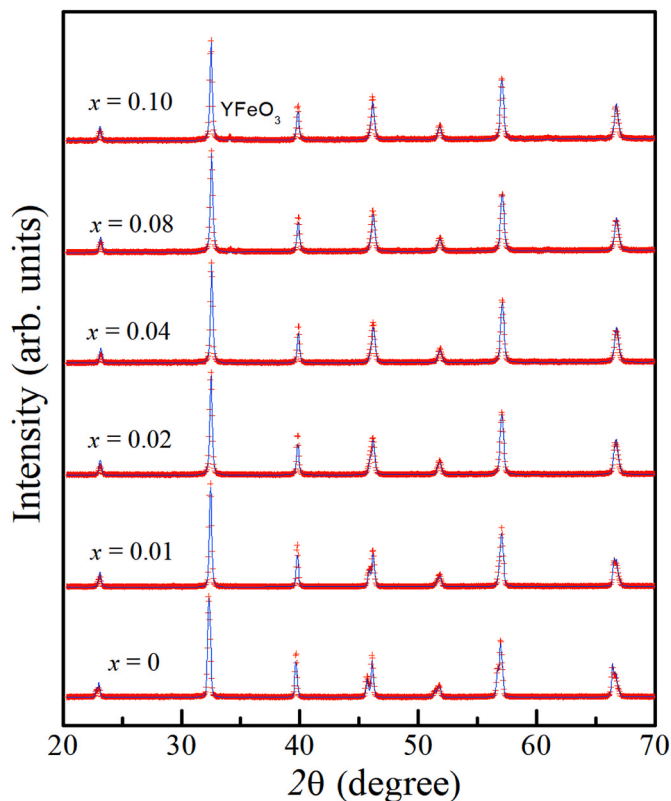


Fig. 2. XRD data of $\text{BY}_x\text{TF}_x\text{O}$ ceramics analyzed as using Rietveld refinement. A small fraction the YFeO_3 -related phase starts constituting as $x = 0.08$.

two samples (corresponding to the middle point of absorption edges) coinciding with the edge of LaFeO_3 . This reflects the presence of Fe^{3+} in $\text{BY}_x\text{TF}_x\text{O}$ ceramics, in good agreement with the charge balance as replacing Y^{3+} (Fe^{3+}) for $\text{Ba}^{2+}(\text{Ti}^{4+})$. As studying (Bi, Fe)- and (Gd, Fe)-codoped BTO materials [41,42], it was also found the presence of Fe^{3+} only. Meanwhile, for Fe-doped BTO materials, there was the mixed

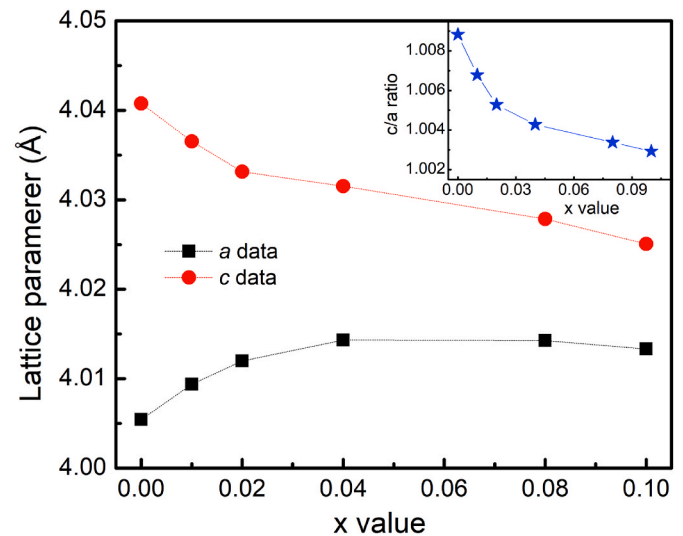


Fig. 3. Variations of the lattice constants, a and c , as a function of x in $\text{BY}_x\text{TF}_x\text{O}$ ceramics. The inset graphs the variation of c/a towards the unity of a cubic structure.

valence of Fe^{3+} and Fe^{4+} [17,33]. Clearly, depending on doping types ($3d$ and/or $4f$ elements), the oxidation state of Fe would be changed.

From the XAS data shown in Fig. 4(a), the analysis of the extended X-ray absorption fine structure (EXAFS) region based on the Fourier transform would provide more the information about local structure around the Fe absorbing atom [44]. Considering EXAFS oscillation spectra of $\text{BY}_x\text{TF}_x\text{O}$ ($x = 0.04$ and 0.1) dependent on the wavevector k , as plotted in Fig. 5(a), one can see that their variation is fairly the same. This proves the samples having the same crystal structure of tetragonality. A small change in vibration intensity as $k = 5\text{--}8 \text{ \AA}^{-1}$ could be associated with locally structural disorder due to the addition of the YFeO_3 secondary phase. If performing Fourier transforms in the R -radial space, with $R = 0\text{--}6 \text{ \AA}$, the obtained spectra show a strong peak around 1.43 and $1.40 \text{ \AA}$ for $x = 0.04$ and 0.1 , respectively, see in Fig. 5(b). This maximum peak is associated with photoelectron scattering at the first

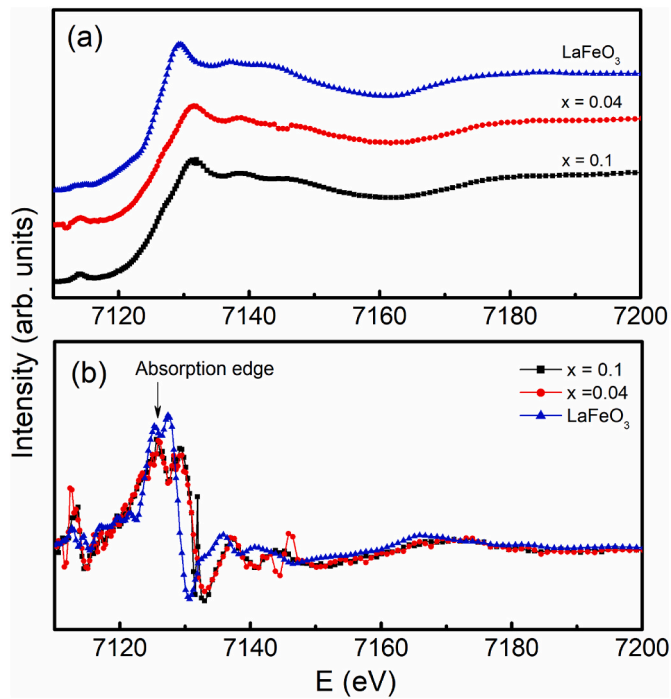


Fig. 4. (a) Fe K-edge XAS and (b) derivative spectra of typical $\text{BY}_x\text{TF}_x\text{O}$ samples with $x = 0.04$ and 0.1 compared with those of LaFeO_3 reference.

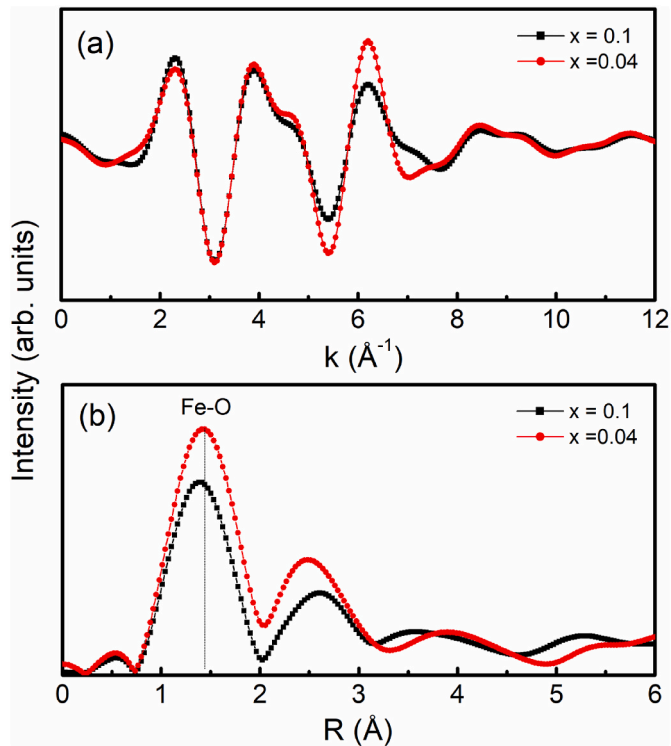


Fig. 5. (a) EXAFS oscillations versus the wavevector k , and (b) Fourier-transformed EXAFS spectra of typical $\text{BY}_x\text{TF}_x\text{O}$ samples with $x = 0.04$ and 0.1 .

shell centering from the Fe absorbing atom, which is assigned to the Fe–O bond distance. Compensating the phase shift (~ 0.5 $\text{\AA}$) due to photoelectron backscattering, the values of Fe–O would be about 1.93 and 1.90 $\text{\AA}$ for $x = 0.04$ and 0.1 , respectively. These value are fairly close to the Ti–O distances (notably, $2 \times (\text{Ti–O}) = c$), proving the Fe^{3+} substitution into the Ti site of tetragonal BTO structure. Thus, the decrease

of Fe–O/Ti–O with increasing x is related to the shrinking of c , in agreement with the change of the lattice parameters towards the cubic structure.

In addition to the morphology and structural analyses, we have assessed phonon modes based on Raman scattering (RS) spectroscopy. Fig. 6 shows RS spectra of all $\text{BY}_x\text{TF}_x\text{O}$ samples. For BTO ($x = 0$), its RS spectrum has vibrational modes peaked at about 199, 261, 311, 519, and 722 cm^{-1} that are assigned to $\text{A}_1(\text{TO})$, $\text{A}_1(\text{TO})$, $\text{E}(\text{TO} + \text{LO})/\text{B}_1$, $\text{A}_1(\text{TO})/\text{E}(\text{TO})$, and $\text{A}_1(\text{LO})/\text{E}(\text{LO})$ characters, respectively, of the tetragonal BTO phase [8,38]. With the (Y, Fe)-codoping in $\text{BY}_x\text{TF}_x\text{O}$ ($x = 0.01$ – 0.1), these modes with a lower intensity still persist. The observed modes around 199, 311 and 519 cm^{-1} associated with the Ti–O asymmetric, TiO_6 -octahedral asymmetric, and O–Ti–O symmetric stretching vibrations, respectively, prove $\text{BY}_x\text{TF}_x\text{O}$ crystallizing in the tetragonal phase. When x increased, more Fe^{3+} ions replaced for Ti^{4+} that would influence the mentioned symmetric and asymmetric vibrations. Consequently, their spectral linewidth has been broadened, and the position of some modes has shifted slightly towards lower frequencies. Here, the intensity suppression of the RS modes versus the Fe^{3+} substitution for Ti^{4+} also suggests the decreased stretching of Ti–O bond distances (*i.e.*, the reduction of c), and tetragonality. Apart from the changes in the intensity and linewidth of the vibration modes, the RS spectra of the samples with $x \geq 0.02$ have the appearance of an additional weak mode peaked at ~ 648 cm^{-1} . Chchiyai et al. [38] assigned it to the distorted perovskite-type structure with the rhombohedral symmetry. With the results obtained from the XRD analysis, we think this mode related to the cubic structure. It means that the Y^{3+} and Fe^{3+} doping has increased the symmetry towards the cubic structure, but no a tetragonal-cubic phase transition like the case of Nd-doped BTO [22] has been observed, probably Fe-doping contents in our samples are still low to the transition. Such structural modifications would impact the magnetic and electric-polarization properties of $\text{BY}_x\text{TF}_x\text{O}$.

Concerning the electric-polarization properties, we have measured

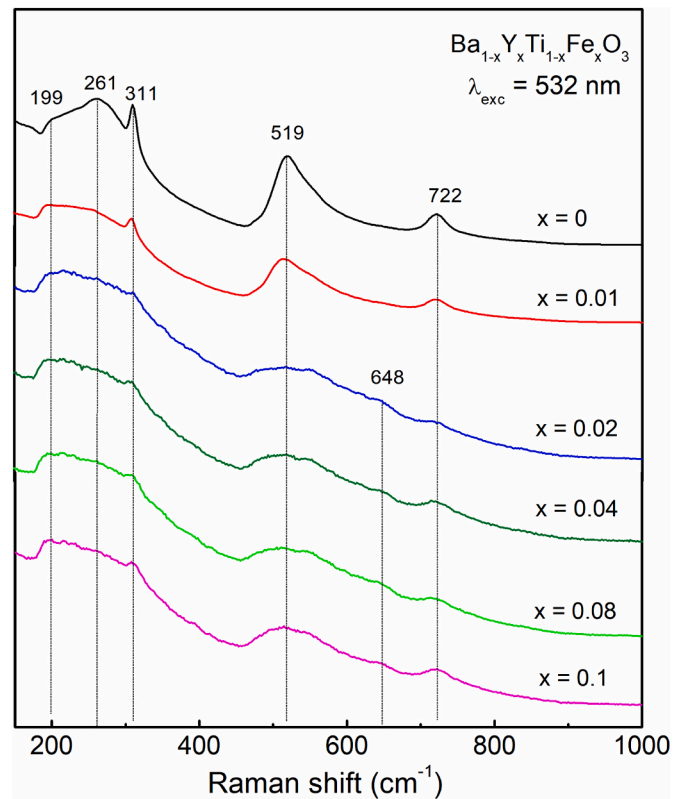


Fig. 6. RS spectra of $\text{BY}_x\text{TF}_x\text{O}$ ceramics excited by a wavelength $\lambda_{\text{exc}} = 532$ nm. The dotted lines correspond to observed vibration modes.

polarization-electric field hysteresis (P - E) loops at room temperature (RT), where $|E|$ could change from 0 to 30 kV/cm. The P - E data graphed in Fig. 7 indicate all the samples exhibiting the ferroelectricity. The maximum polarization ($P_{\max}$) is about $0.051 \mu\text{C}/\text{cm}^2$ for $x = 0$ and gradually decreases through $0.048 \mu\text{C}/\text{cm}^2$ for $x = 0.08$ to $\sim 0.045 \mu\text{C}/\text{cm}^2$ for $x = 0.1$, see Table 1 for more details. Through $|E|$ increases to 30 kV/cm, the hysteresis loops are not saturated, revealing incomplete domain orientation due to cationic disorder [45]. It is known that the spontaneous polarization in BTO is associated with a shift of Ti^{4+} and O^{2-} relative to Ba^{2+} at the origin [46]. The incorporations of Y^{3+} (Fe^{3+}) into the Ba^{2+} (Ti^{4+}) site causes lattice disorders, and dimensional changes towards the cubic structure of BTO, which belongs to the centrosymmetric crystal group with paraelectric and non-piezoelectric characters. Because the above Raman and XRD analyses have demonstrated that the $\text{BY}_x\text{TF}_x\text{O}$ ($x > 0$) samples crystallized in tetragonality with a symmetry higher than that of BTO ($x = 0$), there still exists of non-centrosymmetric regions due to the off-centering of $\text{Ti}^{4+}/\text{Fe}^{3+}$ ions, and possible oxygen vacancies induced by the difference in valence of the replacing ion (Fe^{3+} replaced for Ti^{4+}). As a result, the electric polarization still persists, and $P_{\max}$ decreases with increasing the Y^{3+} and Fe^{3+} content (x) in $\text{BY}_x\text{TF}_x\text{O}$. The same feature was also observed in Fe-doped $\text{Ba}_{0.85}\text{Ca}_{0.15}\text{Ti}_{0.9}\text{Zr}_{0.1}\text{O}_3$ compounds [47]. It is worth noticing that comparing with Ca-doped [48], Mn-doped [49,50], and (Gd, Fe)-codoped BTO materials [42], the $P_{\max}$ values of $\text{BY}_x\text{TF}_x\text{O}$ are smaller. However, if considering the coercive field (E_c) shown in Table 1, its values are fairly large in comparison to E_c of BTO-based compounds. Specifically, E_c decreases from ~ 16.03 kV/cm for $x = 0$ to ~ 13.16 kV/cm for $x = 0.04$, but increases from 13.22 kV/cm for $x = 0.08$ to ~ 15.88 kV/cm as $x = 0.1$. We think that an increase in E_c as $x \geq 0.08$ is also related to the YFeO_3 secondary phase, which influences grain boundaries, and interactions between electric-dipole domains. Furthermore, YFeO_3 was proven to be a ferroelectric material at RT [51] that could partially contribute to the ferroelectricity of $\text{BY}_x\text{TF}_x\text{O}$ as $x \geq 0.08$. The differences in sample preparation/processing conditions, A/B-site doping, and dopant contents have resulted in various microstructures, grain boundaries, porosities, and oxygen vacancies, consequently different $P_{\max}$ and E_c values found in literature.

To learn about the magnetic properties of $\text{BY}_x\text{TF}_x\text{O}$, we have based on a VSM to record field-dependent magnetization (M - H) data. The recorded data at RT plotted in Fig. 8(a) reveal all the samples exhibiting hysteresis phenomenon at $|H|$ values below 5 kOe, which is characteristic of ferromagnetism. Above 5 kOe, both magnetization and

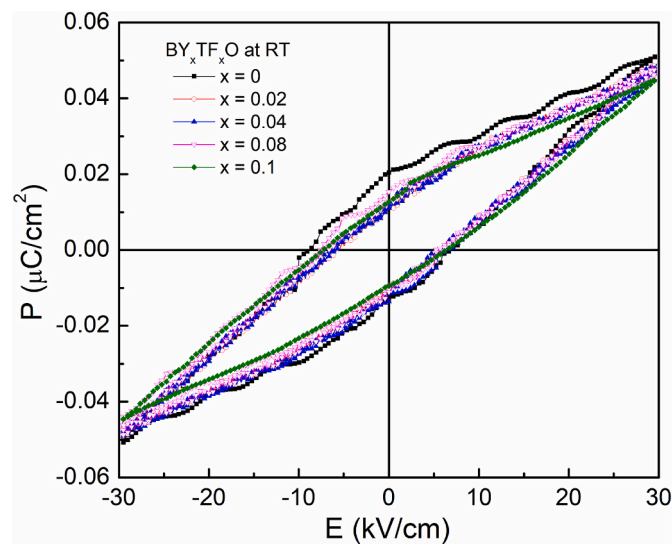


Fig. 7. Polarization-electric field hysteresis loops of typical $\text{BY}_x\text{TF}_x\text{O}$ samples at RT.

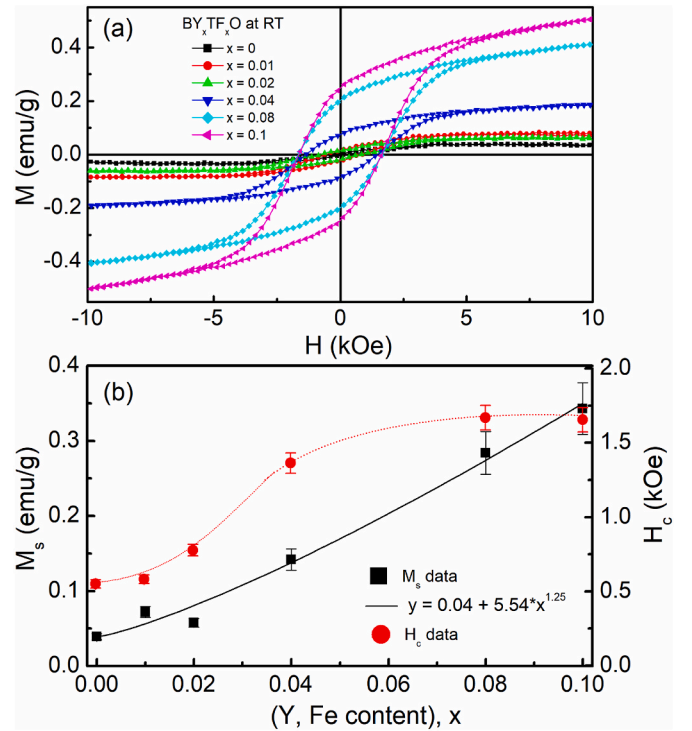


Fig. 8. (a) M - H data of $\text{BY}_x\text{TF}_x\text{O}$ ($x = 0 \sim 0.1$) recorded at RT, and (b) variations of M_s and H_c as a function of x . The dotted line for H_c data is to guide the eyes only.

demagnetization data overlap each other, and M tends to saturate when $|H|$ increases, reflecting the alignment of magnetic moments and domains to the field direction. From the M - H data, we have determined the saturation magnetization (M_s) and coercive force (H_c), as plotted in Fig. 8(b). For H_c , its value ranges from 0.55 to ~ 0.78 kOe for $x = 0 \sim 0.02$, which less changes versus x . However, when x increases from 0.04 to 0.1 , there is a rapid increase of H_c from 1.36 to ~ 1.65 kOe, respectively, see Fig. 8(b). It is known that H_c is dependent on microstructures, grain boundaries, and magnetocrystalline anisotropy. Measurements carried out on the powder samples with the same microstructures would limit the impacts of magnetocrystalline anisotropy. We think that the H_c changes in the samples with $x < 0.08$ are mainly related to magnetic domains formed from exchange interactions between Fe^{3+} ions in tetragonal BTO. Meanwhile, the H_c variation as $x \geq 0.08$ is related to both magnetic domains and YFeO_3 phase. It is necessary to add that YFeO_3 is also a multiferroic material with a large H_c value at RT [51].

Having considered M_s , one can see its gradual increase from ~ 0.04 emu/g for $x = 0$ through 0.14 emu/g for $x = 0.04$ to about 0.34 emu/g for $x = 0.1$. These $M_s(x)$ changes can be described by an analytic function $y = 0.04 + 5.54 \times x^{1.25}$. Temperature measurements at $T = 300 \sim 500$ K for typical $\text{BY}_x\text{TF}_x\text{O}$ samples ($x = 0.01, 0.04$ and 0.1) in an applied field $H = 5$ kOe proclaim no significant change in value of M , see Fig. 9. Similar to M_s , M increases with increasing x , and M of $x = 0.1$ is largest. However, no ferromagnetic-paramagnetic transition has been recorded in the investigated T range. These results prove the magnetic-ordering stability of $\text{BY}_x\text{TF}_x\text{O}$ at high temperatures, above 500 K. Because no magnetic impurities are present in pure BTO ($x = 0$), observed weak ferromagnetism in it could be originated from localized electron spin moments resulting from lattice defects such as oxygen vacancies [15,52] and/or unexpected impurities [53] that were created during sample preparation processes. If giving attention to the $M_s(x)$ variation of the doped samples containing Fe^{3+} ions, as shown in Fig. 8 (b), one can see its change as $x \leq 0.04$ is fairly small, and thus ferromagnetism in these samples is ascribed to oxygen-mediated

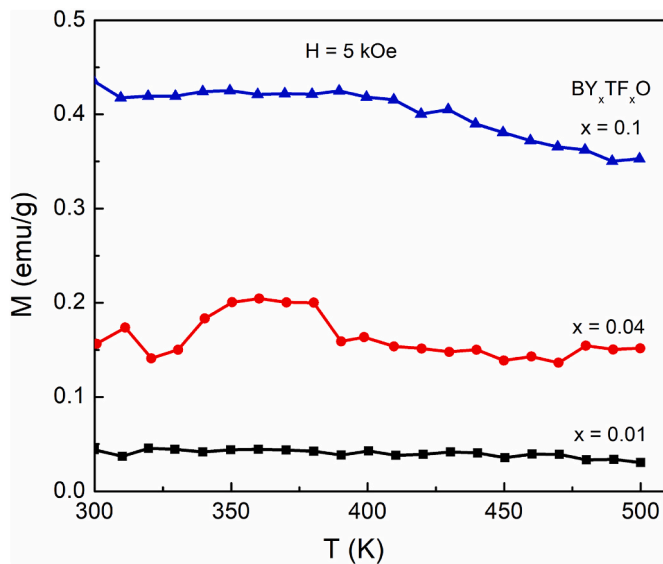


Fig. 9. M - T data of three typical $\text{BY}_x\text{TF}_x\text{O}$ samples ($x = 0.01, 0.04$ and 0.1) in the field $H = 5$ kOe at temperatures $T = 300\sim 500\text{K}$.

superexchange interactions between the Fe^{3+} ions in the BTO lattice. Meanwhile, for the samples $x > 0.04$, M_s varies more rapidly. Combining with the structural analyses, we think that exchange interactions between the Fe^{3+} ions in both BTO and YFeO_3 lattices are responsible to magnetic ordering in these samples, since YFeO_3 is also ferromagnetic above RT [51]. Our results are well consistent with theoretical and experimental studies on Fe^{3+} -doped BTO [19,31,33,54]. Studying $\text{Ba}_{1-x}\text{Gd}_x\text{Ti}_{1-x}\text{Fe}_x\text{O}_3$ ($x = 0.1\sim 0.5$) [42] and (Na, Fe) -codoped BTO [30], it has also been observed ferromagnetism associated with Fe^{3+} ions. However, as investigating $\text{BaTi}_{1-x}\text{Fe}_x\text{O}_3$ with a $\text{Fe}^{3+}/\text{Fe}^{4+}$ mixed valence, it was found either paramagnetic or ferromagnetic behavior, depending on x and crystal structures [8,55]. As reviewing other doped BTO systems, one can see ferromagnetism found in almost $\text{BaTi}_{1-x}\text{Mn}_x\text{O}_3$ [9,40,56–58] and $\text{BaTi}_{1-x}\text{Ni}_x\text{O}_3$ [20] compounds, while paramagnetism and/or ferromagnetism has been found in $\text{BaTi}_{1-x}\text{Co}_x\text{O}_3$ [21,52,59]. These results demonstrate the possibility of successfully fabricating BTO-based multiferroics (showing both ferroelectricity and ferromagnetism) if selecting suitable doping elements and contents, which suggest their application potentials in next multifunctional electronic devices.

4. Conclusions

We have successfully prepared $\text{BY}_x\text{TF}_x\text{O}$ ($x = 0\sim 0.1$) ceramics. All the samples crystallized in a trigonal structure, and a small amount of the YFeO_3 -related secondary phase constituted as $x \geq 0.08$. An increase of x changed the lattice constants (a and c) towards the cubic structure. This reduces the electric polarization, and the Raman intensity of some characteristic modes, due to an increased symmetry. All $\text{BY}_x\text{TF}_x\text{O}$ samples exhibit high-temperature ferromagnetism, and both M_s and H_c increase with increasing x . Based on the results collected as analyzing XAS data, we believe that ferromagnetism in $\text{BY}_x\text{TF}_x\text{O}$ ($x > 0$) is related to super-exchange interactions between Fe^{3+} ions occupying the Ti^{4+} site of the BTO structure. For $x \geq 0.08$, there is an additional contributions of exchange interactions between Fe^{3+} ions in YFeO_3 , which rapidly enhance M_s . These multiferroic properties found in our $\text{BY}_x\text{TF}_x\text{O}$ compounds prove their great potentials in the applications of multifunctional electronic devices.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Acknowledgements

The work was supported by National Research Foundation of Korea (NRF) grants funded by the Korea government's Ministry of Science and ICT (MSIT, 2020R1A2B5B01002184). This work was also supported by Materials, Components & Equipments Research Program funded by the Gyeonggi Province (AICT-003-T1).

References

- [1] Y. Tokura, S. Seki, Multiferroics with spiral spin orders, *Adv. Mater.* 22 (2010) 1554–1565, <https://doi.org/10.1002/adma.200901961>.
- [2] N.A. Spaldin, R. Ramesh, Advances in magnetoelectric multiferroics, *Nat. Mater.* 18 (2019) 203–212, <https://doi.org/10.1038/s41563-018-0275-2>.
- [3] M. Fiebig, Th Lottermoser, D. Fröhlich, A.V. Goltsev, R.V. Pisarev, Observation of coupled magnetic and electric domains, *Nature* 419 (2002) 818–820, <https://doi.org/10.1038/nature01077>.
- [4] C.-W. Nan, N. Cai, L. Liu, J. Zhai, Y. Ye, Y. Lin, Coupled magnetic–electric properties and critical behavior in multiferroic particulate composites, *J. Appl. Phys.* 94 (2003) 5930–5936, <https://doi.org/10.1063/1.1614866>.
- [5] A. Filippetti, N.A. Hill, Coexistence of magnetism and ferroelectricity in perovskites, *Phys. Rev. B* 65 (2002), 195120, <https://doi.org/10.1103/PhysRevB.65.195120>.
- [6] A.M. George, J. Íñiguez, L. Bellaiche, Anomalous properties in ferroelectrics induced by atomic ordering, *Nature* 413 (2001) 54–57, <https://doi.org/10.1038/35092530>.
- [7] U.S. Shenoy, D.K. Bhat, Vanadium-doped BaTiO_3 as high performance thermoelectric material: role of electronic structure engineering, *Mater. Today Chem.* 18 (2020), 100384, <https://doi.org/10.1016/j.mtchem.2020.100384>.
- [8] N.V. Dang, T.D. Thanh, L.V. Hong, V.D. Lam, T.-L. Phan, Structural, optical and magnetic properties of polycrystalline $\text{BaTi}_{1-x}\text{Fe}_x\text{O}_3$ ceramics, *J. Appl. Phys.* 110 (2011), 043914, <https://doi.org/10.1063/1.3625235>.
- [9] T.-L. Phan, P. Zhang, D. Grinting, S.C. Yu, N.X. Nghia, N.V. Dang, V.D. Lam, Influences of annealing temperature on structural characterization and magnetic properties of Mn-doped BaTiO_3 ceramics, *J. Appl. Phys.* 112 (2012), 013909, <https://doi.org/10.1063/1.4733691>.
- [10] H. Zhang, S.W. Or, H.L.W. Chan, Fine-grained multiferroic $\text{BaTiO}_3/(\text{Ni}_0.5\text{Zn}_0.5)\text{Fe}_2\text{O}_4$ composite ceramics synthesized by novel powder-in-sol precursor hybrid processing route, *Mater. Res. Bull.* 44 (2009) 1339–1346, <https://doi.org/10.1016/j.materresbull.2008.12.007>.
- [11] M.P. Singh, W. Prellier, L. Mechin, Ch Simon, B. Raveau, Correlation between structure and properties in multiferroic $\text{La}_0.7\text{Ca}_0.3\text{MnO}_3/\text{BaTiO}_3$ superlattices, *J. Appl. Phys.* 99 (2006), 024105, <https://doi.org/10.1063/1.2161424>.
- [12] M. ud D. Rather, R. Samad, N. Hassan, B. Want, Magnetodielectric effect in rare earth doped $\text{BaTiO}_3\text{-CoFe}_2\text{O}_4$ multiferroic composites, *J. Alloys Compd.* 794 (2019) 402–416, <https://doi.org/10.1016/j.jallcom.2019.04.244>.
- [13] Y. Fo, Y. Ma, H. Dong, X. Zhou, Tuning the electronic structure of BaTiO_3 for an enhanced photocatalytic performance using cation–anion codoping: a first-principles study, *New J. Chem.* 45 (2021) 8228–8239, <https://doi.org/10.1039/D1NJ00283J>.
- [14] M.H. Frey, D.A. Payne, Grain-size effect on structure and phase transformations for barium titanate, *Phys. Rev. B* 54 (1996) 3158–3168, <https://doi.org/10.1103/PhysRevB.54.3158>.
- [15] T.-L. Phan, P. Zhang, D.S. Yang, T.D. Thanh, D.A. Tuan, S.C. Yu, Origin of ferromagnetism in BaTiO_3 nanoparticles prepared by mechanical milling, *J. Appl. Phys.* 113 (2013) 17E305, <https://doi.org/10.1063/1.4799473>.
- [16] Y.-H. Lin, J. Yuan, S. Zhang, Y. Zhang, J. Liu, Y. Wang, C.-W. Nan, Multiferroic behavior observed in highly orientated Mn-doped BaTiO_3 thin films, *Appl. Phys. Lett.* 95 (2009), 033105, <https://doi.org/10.1063/1.3182793>.
- [17] H.M. Nguyen, N.V. Dang, P.-Y. Chuang, T.D. Thanh, C.-W. Hu, T.-Y. Chen, V. D. Lam, C.-H. Lee, L.V. Hong, Tetragonal and hexagonal polymorphs of $\text{BaTi}_{1-x}\text{Fe}_x\text{O}_3$ – δ multiferroics using x-ray and Raman analyses, *Appl. Phys. Lett.* 99 (2011), 202501, <https://doi.org/10.1063/1.3662389>.
- [18] F. Bourguiba, H. Kacem, S. Nasr, M. Abassi, E.K. Hilal, J. Dhahri, K. Taibi, H. Belmabrouk, Electrical and magnetic properties of 6H- BaTiO_3 – δ B site doped: the case of double cation couple ($\text{Fe}^{3+}/\text{Mo}^{6+}$) and ($\text{Fe}^{3+}/\text{W}^{6+}$), *Inorg. Chem. Commun.* 145 (2022), 109882, <https://doi.org/10.1016/j.inoche.2022.109882>.
- [19] M. Staruch, H. Elbidweihy, M.G. Cain, P. Thompson, C.A. Lucas, P. Finkel, Magnetic and multiferroic properties of dilute Fe-doped BaTiO_3 crystals, *Appl. Mater.* 8 (2020), 031109, <https://doi.org/10.1063/5.0002863>.
- [20] S.K. Das, R.N. Mishra, B.K. Roul, Magnetic and ferroelectric properties of Ni doped BaTiO_3 , *Solid State Commun.* 191 (2014) 19–24, <https://doi.org/10.1016/j.ssc.2014.04.001>.
- [21] A. Rani, J. Kolte, P. Gopalan, Structural, electrical, magnetic and magnetoelectric properties of Co-doped BaTiO_3 multiferroic ceramics, *Ceram. Int.* 44 (2018) 16703–16711, <https://doi.org/10.1016/j.ceramint.2018.06.098>.
- [22] Z. Yao, H. Liu, Y. Liu, Z. Wu, Z. Shen, Y. Liu, M. Cao, Structure and dielectric behavior of Nd-doped BaTiO_3 perovskites, *Mater. Chem. Phys.* 109 (2008) 475–481, <https://doi.org/10.1016/j.matchemphys.2007.12.019>.

- [23] K.-J. Park, C.-H. Kim, Y.-J. Yoon, S.-M. Song, Y.-T. Kim, K.-H. Hur, Doping behaviors of dysprosium, yttrium and holmium in BaTiO₃ ceramics, *J. Eur. Ceram. Soc.* 29 (2009) 1735–1741, <https://doi.org/10.1016/j.jeurceramsoc.2008.10.021>.
- [24] S.B. Narang, D. Kaur, S. Bahel, Dielectric properties of lanthanum substituted barium titanate microwave ceramics, *Mater. Lett.* 60 (2006) 3179–3182, <https://doi.org/10.1016/j.matlet.2006.02.079>.
- [25] S. Sahoo, S. Yadav, M. Chandra, K. Singh, Enhancement of Dielectric and Ferroelectric Properties by Ca Doping BaTiO₃, 2019, 030020, <https://doi.org/10.1063/1.5112859>.
- [26] F.D. Morrison, D.C. Sinclair, A.R. West, Doping mechanisms and electrical properties of La-doped BaTiO₃ ceramics, *Int. J. Inorg. Mater.* 3 (2001) 1205–1210, [https://doi.org/10.1016/S1466-6049\(01\)00128-3](https://doi.org/10.1016/S1466-6049(01)00128-3).
- [27] W. Li, J. Hao, W. Bai, Z. Xu, R. Chu, J. Zhai, Enhancement of the temperature stabilities in yttrium doped (Ba_{0.99}Ca_{0.01})(Ti_{0.98}Zr_{0.02})O₃ ceramics, *J. Alloys Compd.* 531 (2012) 46–49, <https://doi.org/10.1016/j.jallcom.2012.03.110>.
- [28] P. Kumar Patel, K.L. Yadav, Effect of yttrium on microstructure, dielectric, ferroelectric and optical properties of BaZr_{0.10}Ti_{0.90}O₃ nanoceramics, *Phys. B Condens. Matter* 442 (2014) 39–43, <https://doi.org/10.1016/j.physb.2014.02.020>.
- [29] M.K. Adak, D. Mondal, S. Mondal, S. Kar, S.J. Mahato, U. Mahato, U.R. Gorai, U. K. Ghorai, D. Dhak, Ferroelectric and photocatalytic behavior of Mn- and Ce-doped BaTiO₃ nanoceramics prepared by chemical route, *Mater. Sci. Eng., B* 262 (2020), 114800, <https://doi.org/10.1016/j.mseb.2020.114800>.
- [30] S.M. Yakout, Influence of Na and Na/Fe doping on the dielectric constant, ferromagnetic and sunlight photocatalytic properties of BaTiO₃ perovskite, *J. Solid State Chem.* 290 (2020), 121517, <https://doi.org/10.1016/j.jssc.2020.121517>.
- [31] D.P. Dutta, M. Roy, N. Maiti, A.K. Tyagi, Phase evolution in sonochemically synthesized Fe³⁺ doped BaTiO₃ nanocrystallites: structural, magnetic and ferroelectric characterisation, *Phys. Chem. Chem. Phys.* 18 (2016) 9758–9769, <https://doi.org/10.1039/C5CP07736B>.
- [32] D. Ginting, S.C. Yu, T.L. Phan, N.V. Dang, T.D. Thanh, V.D. Lam, Electron-spin-resonance spectra and ferroelectricity of BaTi_{1-x}Fe_xO₃, *J. Kor. Phys. Soc.* 62 (2013) 2128–2132, <https://doi.org/10.3938/jkps.62.2128>.
- [33] I.N. Apostolova, A.T. Apostolov, S. Golrokh Bahoosh, J.M. Wesselinowa, Origin of ferromagnetism in transition metal doped BaTiO₃, *J. Appl. Phys.* 113 (2013), 203904, <https://doi.org/10.1063/1.4807412>.
- [34] G.P. Khanal, S. Kim, I. Fujii, S. Ueno, C. Moriyooshi, Y. Kuroiwa, S. Wada, Effect of thermal annealing on crystal structures and electrical properties in BaTiO₃ ceramics, *J. Appl. Phys.* 124 (2018), 034102, <https://doi.org/10.1063/1.5023814>.
- [35] D.L. Kaiser, M.D. Vaudin, L.D. Rotter, Z.L. Wang, J.P. Cline, C.S. Hwang, R. B. Marinenko, J.G. Gillen, Epitaxial growth of BaTiO₃ thin films at 600 °C by metalorganic chemical vapor deposition, *Appl. Phys. Lett.* 66 (1995) 2801–2803, <https://doi.org/10.1063/1.113480>.
- [36] M. Yashima, T. Hoshina, D. Ishimura, S. Kobayashi, W. Nakamura, T. Tsurumi, S. Wada, Size effect on the crystal structure of barium titanate nanoparticles, *J. Appl. Phys.* 98 (2005), 014313, <https://doi.org/10.1063/1.1935132>.
- [37] R.D. Shannon, Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides, *Acta Crystallogr. A* 32 (1976) 751–767, <https://doi.org/10.1107/S0567739476001551>.
- [38] Z. Chchiyai, F. El Bachraoui, Y. Tamraoui, L. Bih, A. Faik, J. Alami, B. Manoun, Design and characterization of novel manganese perovskites Ba_{1-x}BixTi_{1-x}MnxO₃ (0 ≤ x ≤ 0.2), *Ceram. Int.* 46 (2020) 26911–26922, <https://doi.org/10.1016/j.ceramint.2020.07.169>.
- [39] M. Arshad, H. Du, M.S. Javed, A. Maqsood, I. Ashraf, S. Hussain, W. Ma, H. Ran, Fabrication, structure, and frequency-dependent electrical and dielectric properties of Sr-doped BaTiO₃ ceramics, *Ceram. Int.* 46 (2020) 2238–2246, <https://doi.org/10.1016/j.ceramint.2019.09.208>.
- [40] A. Rani, J. Kolte, P. Gopalan, Phase formation, microstructure, electrical and magnetic properties of Mn substituted barium titanate, *Ceram. Int.* 41 (2015) 14057–14063, <https://doi.org/10.1016/j.ceramint.2015.07.023>.
- [41] Z. Chchiyai, F. El Bachraoui, Y. Tamraoui, E.M. Haily, L. Bih, A. Lahmar, J. Alami, B. Manoun, Design, structural evolution, optical, electrical and dielectric properties of perovskite ceramics Ba_{1-x}BixTi_{1-x}Fe_xO₃ (0 ≤ x ≤ 0.8), *Mater. Chem. Phys.* 273 (2021), 125096, <https://doi.org/10.1016/j.matchemphys.2021.125096>.
- [42] S. Sahoo, P.K. Mahapatra, R.N.P. Choudhary, P. Alagarsamy, Influence of compositional variation on structural, electrical and magnetic characteristics of (Ba_{1-x}Gd_x)(Ti_{1-x}Fe_x)O₃ (0.2 ≤ x ≤ 0.5), *Mater. Res. Express* 5 (2018), 016101, <https://doi.org/10.1088/2053-1591/aa9efb>.
- [43] J. Fujioka, A. Doi, D. Okuyama, D. Morikawa, T. Arima, K.N. Okada, Y. Kaneko, T. Fukuda, H. Uchiyama, D. Ishikawa, A.Q.R. Baron, K. Kato, M. Takata, Y. Tokura, Ferroelectric-like metallic state in electron doped BaTiO₃, *Sci. Rep.* 5 (2015), 13207, <https://doi.org/10.1038/srep13207>.
- [44] B.K. Teo, EXAFS: Basic Principles and Data Analysis, Springer Berlin Heidelberg, Berlin, Heidelberg, 1986, <https://doi.org/10.1007/978-3-642-50031-2>.
- [45] P. Esther Rubavathi, L. Venkidu, M. Veera Gajendra Babu, R. Venkat Raman, B. Bagyalakshmi, S.M. Abdul Kader, K. Baskar, M. Muneeswaran, N.V. Giridharan, B. Sundarakannan, Structure, morphology and magnetodielectric investigations of BaTi_{1-x}Fe_xO_{3-δ} ceramics, *J. Mater. Sci. Mater. Electron.* 30 (2019) 5706–5717, <https://doi.org/10.1007/s10854-019-00864-6>.
- [46] M. Acosta, N. Novak, V. Rojas, S. Patel, R. Vaish, J. Koruza, G.A. Rossetti, J. Rödel, BaTiO₃-based piezoelectrics: fundamentals, current status, and perspectives, *Appl. Phys. Rev.* 4 (2017), 041305, <https://doi.org/10.1063/1.4990046>.
- [47] E. Venkata Ramana, F. Figueiras, A. Mahajan, D.M. Tobaldi, B.F.O. Costa, M.P. F. Graça, M.A. Valente, Effect of Fe-doping on the structure and magnetoelectric properties of (Ba_{0.85}Ca_{0.15})(Ti_{0.9}Zr_{0.1})O₃ synthesized by a chemical route, *J. Mater. Chem. C* 4 (2016) 1066–1079, <https://doi.org/10.1039/C5TC00914F>.
- [48] R. Muhammad, M.A. Khalil, M.S. Castro, Structure and dielectric characteristics of Ba_{1-x}Ca_xTi_{1-x}O_{3-δ} ceramics, *Ceram. Int.* 46 (2020) 1059–1064, <https://doi.org/10.1016/j.ceramint.2019.09.072>.
- [49] Y. Ichikawa, Y. Kitanaka, T. Oguchi, Y. Noguchi, M. Miyayama, Polarization degradation and oxygen-vacancy rearrangement in Mn-doped BaTiO₃ ferroelectrics ceramics, *J. Ceram. Soc. Jpn.* 122 (2014) 373–380, <https://doi.org/10.2109/jcersj2.122.373>.
- [50] W. Chen, X. Zhao, J. Sun, L. Zhang, L. Zhong, Effect of the Mn doping concentration on the dielectric and ferroelectric properties of different routes-fabricated BaTiO₃-based ceramics, *J. Alloys Compd.* 670 (2016) 48–54, <https://doi.org/10.1016/j.jallcom.2016.01.187>.
- [51] M. Shang, C. Zhang, T. Zhang, L. Yuan, L. Ge, H. Yuan, S. Feng, The multiferroic perovskite YFeO₃, *Appl. Phys. Lett.* 102 (2013), 062903, <https://doi.org/10.1063/1.4791697>.
- [52] L. Yang, H. Qiu, L. Pan, Z. Guo, M. Xu, J. Yin, X. Zhao, Magnetic properties of BaTiO₃ and BaTi_{1-x}MxO₃ (M=Co, Fe) nanocrystals by hydrothermal method, *J. Magn. Magn. Mater.* 350 (2014) 1–5, <https://doi.org/10.1016/j.jmmm.2013.09.036>.
- [53] G. Alvarez, A. Conde-Gallardo, H. Montiel, R. Zamorano, About room temperature ferromagnetic behavior in BaTiO₃ perovskite, *J. Magn. Magn. Mater.* 401 (2016) 196–199, <https://doi.org/10.1016/j.jmmm.2015.10.031>.
- [54] A. Zorko, M. Pregelj, M. Gomilšek, Z. Jagličić, D. Pajić, M. Telling, I. Arcon, I. Mikulska, M. Valant, Strain-induced extrinsic high-temperature ferromagnetism in the Fe-doped hexagonal barium titanate, *Sci. Rep.* 5 (2015) 7703, <https://doi.org/10.1038/srep07703>.
- [55] L. Zhou, Y. Zhang, S. Li, Q. Lian, J. Yang, W. Bai, X. Tang, Fe doping effect on the structural, ferroelectric and magnetic properties of polycrystalline BaTi_{1-x}Fe_xO₃ ceramics, *J. Mater. Sci. Mater. Electron.* 31 (2020) 14487–14493, <https://doi.org/10.1007/s10854-020-04008-z>.
- [56] N.V. Dang, N.T. Dang, T.A. Ho, N. Tran, T.L. Phan, Electronic structure and magnetic properties of BaTi_{1-x}Mn_xO₃, *Curr. Appl. Phys.* 18 (2018) 150–154, <https://doi.org/10.1016/j.cap.2017.11.020>.
- [57] K. Madhan, C. Jagadeeshwaran, R. Murugaraj, Enhancement of electrical and magnetic properties in acceptor-doped BaTiO₃ ferroelectric ceramics, *J. Mater. Sci. Mater. Electron.* 30 (2019) 2953–2965, <https://doi.org/10.1007/s10854-018-00573-6>.
- [58] N.V. Dang, T.-L. Phan, T.D. Thanh, V.D. Lam, L.V. Hong, Structural phase separation and optical and magnetic properties of BaTi_{1-x}Mn_xO₃ multiferroics, *J. Appl. Phys.* 111 (2012), 113913, <https://doi.org/10.1063/1.4725195>.
- [59] T.-L. Phan, P.D. Thang, T.A. Ho, T.V. Manh, T.D. Thanh, V.D. Lam, N.T. Dang, S. C. Yu, Local geometric and electronic structures and origin of magnetism in Co-doped BaTiO₃ multiferroics, *J. Appl. Phys.* 117 (2015) 17D904, <https://doi.org/10.1063/1.4907182>.